

# claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

## Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a1ff7d8c9074df20f.jsonl`  
- SHA-256: `7278f1e2578e23dd1f15653de61fc3b9334a3c4438a05e9d89624a2266d28fa8`  
- Source modified: `2026-05-31T04:26:20+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

## Transcript

1. user (2026-05-31T04:25:59.839Z)

### Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

### Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract\_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

## Claim under review

"OmniDocBench is a CVPR 2025 document-parsing benchmark by OpenDataLab (Shanghai AI Lab) covering 1,651 PDF pages across 10 document types ? and it explicitly includes financial reports as one of the document categories, making its table-recognition results directly relevant to bank-statement parsing."

**\*\*Source:\*\*** <https://github.com/opendatalab/OmniDocBench> (primary)

**\*\*Supporting quote:\*\*** "It encompasses a wide range of content, including academic papers, financial reports, newspapers, textbooks, and handwritten notes."

## Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

**\*\*refuted=true\*\*** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

**\*\*refuted=false\*\*** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

### 2. assistant (2026-05-31T04:26:02.367Z)

I'll verify this claim about OmniDocBench. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

### 3. assistant (2026-05-31T04:26:05.364Z)

I have the tools loaded. Let me investigate this claim about OmniDocBench by fetching the primary source and searching for corroborating evidence.

### 4. user (2026-05-31T04:26:12.483Z)

# OmniDocBench Overview

## **\*\*Page Count & Document Types:\*\***

OmniDocBench contains "1651 PDF pages, covering 10 document types, 5 layout types, and 5 language types."

## **\*\*Confirmed Document Categories:\*\***

The benchmark includes academic papers, financial reports, newspapers, textbooks, and handwritten notes. The full list encompasses: academic literature, PPT-to-PDF, books, colorful textbooks, exam papers, handwritten notes, magazines, research reports, and newspapers.

## **\*\*CVPR 2025 Status:\*\***

Yes, the project states "OmniDocBench has been accepted by CVPR 2025!"

## **\*\*Creator:\*\***

OpenDataLab developed OmniDocBench. The GitHub repository is maintained by the opendatalab organization, which is affiliated with Shanghai AI Lab.

## **\*\*Financial Reports Confirmation:\*\***

The documentation explicitly mentions "financial reports" as a covered category, appearing in descriptions of document types and data source classifications.

## **5. user (2026-05-31T04:26:13.703Z)**

Web search results for query: "OmniDocBench CVPR 2025 document types categories 1651 pages financial reports"

Links: [{"title": "GitHub - marklabz/omnidocbench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/marklabz/omnidocbench"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench?tab=readme-ov-file"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "CVPR 2025 Open Access Repository", "url": "https://openaccess.thecvf.com/content/CVPR2025/html/Ouyang\_OmniDocBench\_Benchmarking\_Diverse\_PDF\_Document\_Parsing\_with\_Comprehensive\_Annotations\_CVPR\_2025\_paper.html"}, {"title": "OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA", "url": "https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"}, {"title": "Sumit on X: \\\"OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations Introduces a benchmark for automated document content extraction with annotations across 9 diverse document types and 19 layout categories.\\\" https://t.co/42tsAU9b9a", "url": "https://t.co/HKmcrlqZn\\\" / X", "url": "https://x.com/\_reachsumit/status/1866700755685740991"}, {"title": "CVPR Poster OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations", "url": "https://cvpr.thecvf.com/virtual/2025/poster/34400"}, {"title": "OmniDocBench is Saturated, What's Next for OCR Benchmarks?", "url": "https://www.llamaindex.ai/blog/omnidocbench-is-saturated-what-s-next-for-ocr-benchmarks"}, {"title": "OmniDocBench: Benchmarking Diverse PDF Document Parsing with", "url": "https://openaccess.thecvf.com/content/CVPR2025/papers/Ouyang\_OmniDocBench\_Benchmarking\_Diverse\_PDF\_Document\_Parsing\_with\_Comprehensive\_Annotations\_CVPR\_2025\_paper.pdf"}]

Based on the search results, here's what I found about OmniDocBench presented at CVPR 2025:

## **\*\*Dataset Size and Document Categories:\*\***

OmniDocBench includes 1651 PDF pages, covering 10 document types, 5 layout types, and 5 language types.

## **\*\*Document Types:\*\***

The benchmark covers 9 diverse document categories including academic papers, textbooks, slides, financial reports, newspapers, handwritten notes, exam papers, magazines, and research reports.

## **\*\*Annotations:\*\***

OmniDocBench features rich annotations, containing 28 block-level annotations (text paragraphs,

headings, tables, etc.) and 4 span-level annotations (text lines, inline formulas, subscripts, etc.).

**\*\*Evaluation Focus:\*\***

The benchmark provides corresponding metric calculations for four major modules: text, tables, formulas, and reading order.

OmniDocBench has been accepted by CVPR 2025!

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

#### **6. assistant (2026-05-31T04:26:17.223Z)**

The claim is strongly corroborated so far. Let me verify the exact wording on the primary source and the CVPR paper to confirm the page count, the "financial reports" / "research reports" category naming, and the creator. I want to be precise about whether the category is literally "financial reports" since one source listed "research reports" instead.

#### **7. user (2026-05-31T04:26:19.543Z)**

You've hit your session limit · resets 1:30pm (Asia/Calcutta)

#### **8. user (2026-05-31T04:26:20.136Z)**

Web search results for query: "OmniDocBench document types list "financial report" OR "research report" academic newspaper textbook magazine"

You've hit your session limit · resets 1:30pm (Asia/Calcutta)

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

#### **9. assistant (2026-05-31T04:26:20.563Z)**

You've hit your session limit · resets 1:30pm (Asia/Calcutta)